

Q5) BEGIN

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DECLARE zoneA_capacity, zoneB_capacity, zoneC_capacity = INTEGER
DECLARE zoneA_occupied, zoneB_occupied, zoneC_occupied = INTEGER
DECLARE parked_cars, parked_bikes, parked_vans = INTEGER
DECLARE rejected_vehicles, accepted_vehicles = INTEGER
DECLARE total_vehicles, vehicle = INTEGER
DECLARE required_spaces = INTEGER
DECLARE vehicle_type, category, permit, emergency = CHARACTER
DECLARE assigned_zone = STRING
DECLARE parked = BOOLEAN

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zoneA_capacity ← 20, zoneB_capacity ← 40
zoneC_capacity ← 15, zoneA_occupied ← 0
zoneB_occupied ← 0, zoneC_occupied ← 0
parked_cars ← 0, parked_bikes ← 0
parked_vans ← 0, rejected_vehicles ← 0

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PRINT "Enter number of vehicles expected:"

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INPUT total_vehicles

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FOR vehicle ← 1 TO total_vehicles

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    PRINT "Vehicle", vehicle

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    REPEAT

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        PRINT "Enter vehicle type (C=Car, B=Bike, V=Van):"

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        INPUT vehicle_type

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        IF vehicle_type <> C AND vehicle_type <> B AND
           vehicle_type <> V THEN

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            PRINT "Invalid vehicle type. Try again"

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        ENDIF

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    UNTIL vehicle_type = C OR vehicle_type = B OR vehicle_type = V

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REPEAT

PRINT "Enter user category (F=faculty, S=Student, G=Visitor):"

INPUT category

IF category <> F AND category <> S AND category <> G THEN

PRINT "Invalid category. Try again"

ENDIF

UNTIL category = F OR category = S OR category = G

REPEAT

PRINT "Does the vehicle have a valid permit? (Y/N):"

INPUT permit

IF permit <> Y AND permit <> N THEN

PRINT "Invalid permit value. Try again"

ENDIF

UNTIL permit = Y OR permit = N

IF permit = N THEN

REPEAT

PRINT "Is this an emergency vehicle? (Y/N):"

INPUT emergency

IF emergency <> Y AND emergency <> N THEN

PRINT "Invalid value. Try again"

ENDIF

UNTIL emergency = Y OR emergency = N

ELSE

emergency ← N

ENDIF

parked ← FALSE

assigned_zone ← ""

IF vehicle_type = V THEN

required_spaces ← 2

ELSE

required_spaces $\leftarrow$ 1

Date:

ENDIF

IF permit = N AND emergency = N THEN

PRINT "Rejected: Invalid permit"

rejected_vehicles $\leftarrow$ rejected_vehicles + 1

ELSE

IF category = F THEN

IF zoneA_capacity - zoneA_occupied $\geq$ required_spaces THEN

zoneA_occupied $\leftarrow$ zoneA_occupied + required_spaces

assigned_zone $\leftarrow$ A

parked $\leftarrow$ TRUE

ELSE

print "Rejected: No available space in zone A"

ENDIF

ELSE IF category = S THEN

IF vehicle_type = V THEN

IF zoneC_capacity - zoneC_occupied $\geq$ 2 THEN

zoneC_occupied $\leftarrow$ zoneC_occupied + 2

assigned_zone $\leftarrow$ C

parked $\leftarrow$ TRUE

ELSE

PRINT "Rejected: No space in ~~zone B~~ or zone C"

ENDIF

ELSE IF zoneB_capacity - zoneB_occupied $\geq$ required_spaces THEN

zoneB_occupied $\leftarrow$ zoneB_occupied + required_spaces

assigned_zone $\leftarrow$ B

parked $\leftarrow$ TRUE

ELSE

PRINT "Rejected: No available space in Zone B"

ENDIF

ELSE IF category = C THEN

IF zoneC_capacity - zoneC_occupied $\geq$ required_spaces THEN

zoneC_occupied $\leftarrow$ zoneC_occupied + required_spaces

assigned_zone $\leftarrow$ C

parked $\leftarrow$ TRUE

ELSE

PRINT "Rejected: No available space in Zone C"

ENDIF

ENDIF

IF parked = TRUE THEN

PRINT "Vehicle accepted"

PRINT "Assigned_zone: ", assigned_zone

IF assigned_zone = A THEN

PRINT "Remaining_capacity: ", zoneA_capacity - zoneA_occupied

ELSE IF assigned_zone = B THEN

PRINT "Remaining_capacity: ", zoneB_capacity - zoneB_occupied

ELSE

PRINT "Remaining capacity: ", zoneC_capacity - zoneC_occupied

ENDIF

IF vehicle_type = C THEN

parked_cars $\leftarrow$ parked_cars + 1

ELSE IF vehicle_type = B THEN

parked_bikes $\leftarrow$ parked_bikes + 1

ELSE

parked_vans $\leftarrow$ parked_vans + 1

ENDIF

ELSE

rejected_vehicles $\leftarrow$ rejected_vehicles + 1

ENDIF
ENDIF

NEXT vehicle

accepted_vehicles ← parked_cars + parked_bikes + parked_vans

PRINT "Total vehicles processed: ", total_vehicles

PRINT "Total accepted vehicles: ", accepted_vehicles

PRINT "Total rejected vehicles: ", rejected_vehicles

PRINT "Cars successfully parked: ", parked_cars

PRINT "Bikes successfully parked: ", parked_bikes

PRINT "Vans successfully parked: ", parked_vans

PRINT "Zone A occupancy: ", zoneA_occupied

PRINT "Zone A remaining capacity: ", zoneA_capacity - zoneA_occupied

PRINT "Zone B occupancy: ", zoneB_occupied

PRINT "Zone B remaining capacity: ", zoneB_capacity - zoneB_occupied

PRINT "Zone C occupancy: ", zoneC_occupied

PRINT "Zone C remaining capacity: ", zoneC_capacity - zoneC_occupied

IF zoneA_occupied ≥ zoneB_occupied AND zoneA_occupied ≥ zoneC_occupied THEN

PRINT "Zone with highest occupancy: Zone A"

ELSE IF zoneB_occupied ≥ zoneA_occupied AND zoneB_occupied ≥ zoneC_occupied THEN

PRINT "Zone with highest occupancy: Zone B"

ELSE

PRINT "Zone with highest occupancy: Zone C"

ENDIF

IF zoneA_occupied = zoneA_capacity AND zoneB_occupied = zoneB_capacity AND zoneC_occupied = zoneC_capacity THEN

PRINT "Entire campus parking facility is full"

ELSE PRINT "Entire campus parking facility is NOT full"

END IF